Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic AI · LangGraph · LLMs · RAG Pipelines · Prompt Engineering · Python · Production AI

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PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production engineering and hands-on expertise in **Agentic AI, LangGraph, LLMs, and RAG** — designing and deploying autonomous AI agents that perform complex tasks reliably in production. Built **Leasel**: an 8-agent LangGraph StateGraph where autonomous agents (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) work end-to-end without human intervention except at confidence-gated decision points — 93.6% precision across 1,240 documents, CI-integrated LLM-as-judge eval, zero production safety violations. Shipped Entrata's first production AI pipeline across 1.6M financial records. Expert in LangGraph/LangChain agent workflows, prompt engineering (few-shot, chain-of-thought, role specification), RAG pipeline design, and integrating LLMs with enterprise APIs. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic AI & Autonomous Agents | LangGraph & LangChain | LLM Integration (GPT, Claude, Gemini, Llama) | RAG Pipeline Design | Prompt Engineering | AI Workflow Orchestration | Multi-Agent Systems | Python & REST API Integration | LLM Evaluation & LLMOps | Production AI & Reliability | HITL & Agent Governance | Cross-Functional AI Delivery

EXPERIENCE

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Python · Statistical Modeling · Instruction

- Built AI/ML prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection to automate data integrity checks; designed analytics dashboards improving research comprehension by 40% across two faculty cohorts.
- Mentored 600+ students across Python programming (data structures, OOP, automation) and database design (SQL, ER modeling, normalization).

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Production AI · Agentic Workflows · Python · API Integration

- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) — designed autonomous data processing workflows with output validation gates, quality controls, and compliance enforcement; zero violations at launch.
- **Integrated LLMs and AI models with enterprise systems via REST APIs** — connected AI outputs to production data pipelines, compliance modules, and reporting systems; built and consumed 20+ APIs across Salesforce, AWS, and internal services; ensured reliability and scalability across 100K+ monthly active users.
- **Built and optimized AI agent behavior in production**: designed troubleshooting and observability tooling reducing debugging time 40%; maintained production bug count below 10 for 6+ consecutive months; optimized 200+ SQL queries (30% gain) across complex production schemas.
- Eliminated 80% of manual operational workload via **Python automation workflows**; led 120+ Agile sprints collaborating cross-functionally with Compliance, Legal, Operations, and Product to deliver AI-driven solutions; achieved 100% PHI compliance on AWS S3 migration (zero security incidents).

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate Software Engineer — Python · Backend · Data Pipelines

- Built backend compliance modules (Python, PHP, PostgreSQL); processed 1.6M HUD/Tax Credit records; migrated client data to AWS S3; contributed to API integrations and database architecture redesigns.

PORTFOLIO PROJECTS

Leasel — 8-Agent Autonomous AI System (LangGraph, RAG, HITL) Agentic AI LangGraph RAG

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, TF-IDF RAG, OCR, SQLite, OpenAI/Anthropic APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed and built a production-grade **autonomous AI agent system**: 8-agent LangGraph StateGraph where agents independently perform triage, retrieval, configuration analysis, historical pattern matching, structured data querying, reasoning, validation, and conditional human escalation — each agent operates autonomously with tool use, function calling, and inter-agent state handoffs; no human in the loop except at confidence-gated regulated decision points.
- **RAG pipeline** with TF-IDF retrieval and 100% citation correctness; **prompt engineering** throughout (chain-of-thought, few-shot, role specification, XML-structured outputs); LLMs integrated via REST APIs (OpenAI, Anthropic) with multi-provider routing and output validation; **optimized agent performance and reliability** via CI-integrated LLM-as-judge evaluation (multi-trial sampling, regression gate) — regressions blocked before merge.
- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs (5 US states + federal); 0 safety boundary violations on full adversarial validation set; production-grade scalability via FastAPI REST endpoints.

AI on FHIR — LLM-Powered Enterprise API Integration LLM Integration REST APIs Agentic Workflow

2024 · Flask, LLMs, FHIR APIs, React, Docker, spaCy, BERT, Embeddings · github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual data retrieval time ~60% by building an AI workflow integrating LLMs + BERT to autonomously translate natural language queries into FHIR REST API calls over enterprise medical records — HIPAA-compliant RBAC; multi-language support (EN/ES); deployed on Docker with full enterprise API integration.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Agentic AI & LLMs: Agentic AI, Autonomous AI Agents, Multi-Agent Orchestration, LangGraph, LangChain, LLM Orchestration, Tool Calling, Function Calling, AI Workflows, LLMs (GPT, Claude, Gemini, Llama), OpenAI API, Anthropic API, Groq, Ollama, Prompt Engineering, Few-Shot Prompting, Zero-Shot Prompting, Chain-of-Thought Prompting, Role Specification, Output Validation, Hallucination Detection, HITL, Confidence Scoring, Guardrails

RAG & Retrieval: RAG Pipelines, Retrieval-Augmented Generation, Hybrid Retrieval, Semantic Search, Embeddings, Vector Database, ChromaDB, TF-IDF, NLP, spaCy, BERT, Hugging Face, OCR, Document Intelligence

LLMOps & Evaluation: LLMOps, LLM Evaluation, LLM-as-Judge, CI-Integrated Evals, Agent Observability, Agent Tracing, Drift Detection, Regression Testing, Model Evaluation

Backend & Infrastructure: Python (primary), FastAPI, REST APIs, Flask, React, TypeScript, JavaScript, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, SQL, RabbitMQ, Microservices, Git, Linux, Agile/Scrum